

mer switch. Jitin Ranjit, managing partner, Enlite Energy Solutions informs, "The standard does not require any special devices to coordinate with the lighting network, making it both easy and intuitive for consumers to use every day." He adds, "Plus, it makes adding or removing products to the lighting network quick and easy."

"Technologies such as Wi-Fi and Bluetooth LE do not allow the creation of mesh networks in comparison with ZigBee. With ZigBee, one can have single-point control over an entire building," informs Nishant Bhaskar, digital application engineer, Texas Instruments India. He adds, "People are trying to design gateways such as a Wi-Fi to ZigBee gateway, which facilitates total control over a network of lights via an Android phone, which is actually on ZigBee. People are experimenting and merging different technologies for smart lighting."

"So far, ZigBee light link is highly used for smart lighting for the prime reason that it is a worldwide-accepted standard, specifically for lighting," says Thawani.

Efforts to standardise

The lighting industry requires a consolidated proposition on the implementation of the kind of wireless technologies that they implement. The lack of an integrated approach leads to slower market adoption, confusion, incompatible products and could also limit the energy-saving potential. Smart meters, which fall under the category of wireless-controlled connected devices, are increasingly being adopted for metering applications. When smart lighting systems become part of these connected systems that involve smart meters, these will have to be compatible with the wireless connectivity technologies used by smart meters.

In order to deal with the issues above, the connected lighting alliance (TCLA is an alliance founded by industry majors such as Toshiba, GE Lighting, OSRAM, Philips, Panasonic and Lutron to promote a unified voice to convey lighting industry requirements) aims to support open standards, there-

Four important elements of smart lighting systems

LEDs. From traffic signals and mobile phones to televisions and computers, LEDs are widely used for their low-energy use, long lifespan and high-quality light. These are based on semiconductor technology that is now being widely adopted for general illumination. LEDs are also inherently controllable and one of the important components of a smart lighting system.

Software. A crucial aspect of a smart lighting system is a control system that integrates software based regulation to determine when and where to supply light. This kind of intelligence has a great effect on driving-down energy consumption, ensuring that lights are consuming power only when needed.

Sensors. Smart lighting systems also use sensors as a mechanism for ensuring efficient light delivery. These devices provide the capability to distinguish occupancy or other environmental parameters and, accordingly, modify the delivery of light as per pre-defined software instructions. For instance, a software rule could be like 'Is the area occupied? Then, turn the lights on in that space—either all at once or in a sequence as the occupancy progresses—to the desired light level (which, of course, can vary as per requirements).' Another example could be 'Are there other sources of light present? Then, decrease the amount of light being delivered.'

Wireless communication. The power to control lighting from remote locations to cut down on energy consumption, create an atmosphere or just deliver less light is invaluable. Be it at home or office, wireless technologies are the most appropriate for smart lighting. Bi-directional communication, with a centralised control device that allows LED fixtures to 'talk' with each other, is another key element of a smart lighting system. Apart from saving money, time and installation effort, it opens up many alternatives and enables flexibility in terms of positioning of LED fixtures.

However, the key to maximising efficiency is to have networking, intelligence, LEDs and sensing, all fused into a single system.

by enabling global adoption and rise of wireless lighting solutions. The alliance also motivates the development of a smart lighting environment comprising component suppliers, solution providers, compliance testers and standard development organisations.

Security needs to become tighter

Smart lighting systems combine networking and software based intelligence to reap the energy-efficiency benefits of LED based light fixtures, but they can have security concerns too. Smart LED light bulbs, controlled remotely by mobile devices, can be vulnerable to hacking, unless powered by strong encryption algorithms.

Context Information Security recently discovered a security flaw in LIFX smart LED light. By posing as a new bulb joining the network, access could be gained to the master bulb to control every other light bulb connected to the network, thereby exposing the wireless network's credentials. However, this task was not easy, probably due to the complexity of the device and all the reverse engineering it took to hack it. The company, LIFX, had acknowledged

and identified the vulnerability and released a software update to fix the problem.

Needless to say, security needs to be highly prioritised, especially for devices getting onto the IoT arena, before organisations start to connect mission-critical devices and systems.

Improvements in LED technology

There have been key developments on the LED manufacturing side, such as superior materials, enhanced equipment and improved processes that have allowed the latest generation of LEDs to yield admirable light output, visual performance and space savings.

Advanced equipment. In the last decade or so, the equipment used to manufacture LED dyes has experi-



On top is a standard LED, and the bottom of the image shows a high-power LED that provides approximately three-times-higher light output and in a package that is half the size of a standard LED's package